COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions (state changes produce notifications; confirmation writes a ledger entry and a notification) and a role-separated workflow. The application uses Next.js, TypeScript, Prisma and SQLite. GPS, scanning, telephony, payment and carrier integration are outside scope. Deployment is local-only, serving a single warehouse and the four roles above; data integrity is enforced by transactional guards — stock checks and one confirmation transaction (§5).

2. Development process

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Project selected	Scope, responsibilities and contracts recorded (increment register)
17 Oct 2026 test-led review	Order and stock functions complete in development	Defects recorded in the increment register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Review found a failing confirmation endpoint and a missing stock ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	All four role workflows runnable together	Customer, staff and admin workflows verified in the 11 Nov regression (browser checks)
Driver and privacy, 2 Nov 2026	Driver workflow and data minimisation outstanding	Driver assignment and data-minimisation checks pass (tests)
Documentation hardening, 10–11 Nov 2026	Review found diagrams and traceability diverging from the implementation	Diagrams and traceability aligned (traceability)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date CSV export in three actions; regression passed
Availability tests, 11–12 Nov 2026	CI health checks had no regression test	Two regression tests added; suite reached 38 (unit.log)

3. Requirements

Functional requirements and verification

FR	The system shall	Verification
FR-01	create orders with product lines and totals	Test — creation and stock
FR-02	enforce lifecycle and stock reservation or release	Test — lifecycle and stock
FR-03	number orders and shipments and show owner tracking (the staff member or driver currently holding each order)	Test — identifiers and tracking
FR-04	show inventory, replenishment, ledger (inventory movement log) and low-stock alerts	Test — inventory and alerts
FR-05	record event notifications and allow reading	Test — notifications and reads
FR-06	provide admin aggregates and CSV export	Test — aggregates and export
FR-07	authenticate and enforce role and object access	Test — authentication and boundaries
FR-08	assign shipped orders and simulate delivery	Test — assignment and delivery

Each suite runs green in the recorded regression (§6).

NFR-01 performance (API p95 ≤2 s, first contentful paint ≤3 s); NFR-02 CI health (green build, test and browser runs recorded per increment); NFR-03 least privilege and data minimisation — role gates on every mutating endpoint; staff order responses omit customer identity and delivery address (privacy tests); NFR-04 scripted operation effort — every core operation completes in at most three actions, one interface interaction each (walkthrough, §8); NFR-05 process quality — CI-verified increments (workflow).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	Increment register	scheduling plan
RK-02 contract rework	Transition tests pin the FR-02 contract (NFR-05)	transition tests
RK-03 test-week capacity	Priority register	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	Seed regression test covers FR-01/04/06	seed regression test
RK-06 assignments and test-week squeeze	Risk plan	assessment schedule
RK-07 stack unfamiliarity	Setup guide and CI lower ramp-up cost	setup guide
RK-08 report and video compete for time	Risk plan	report and video plan

4. System models

The four layers — Role-specific pages, HTTP routes and authentication, Order/inventory/driver business handlers, Prisma/SQLite persistence — each isolate one concern. Figures 1–5 answer, in order: which roles perform each function; which transitions and stock effects are valid; how items, orders, shipments and inventory logs relate; how confirmation commits or rolls back; how the four layers connect.

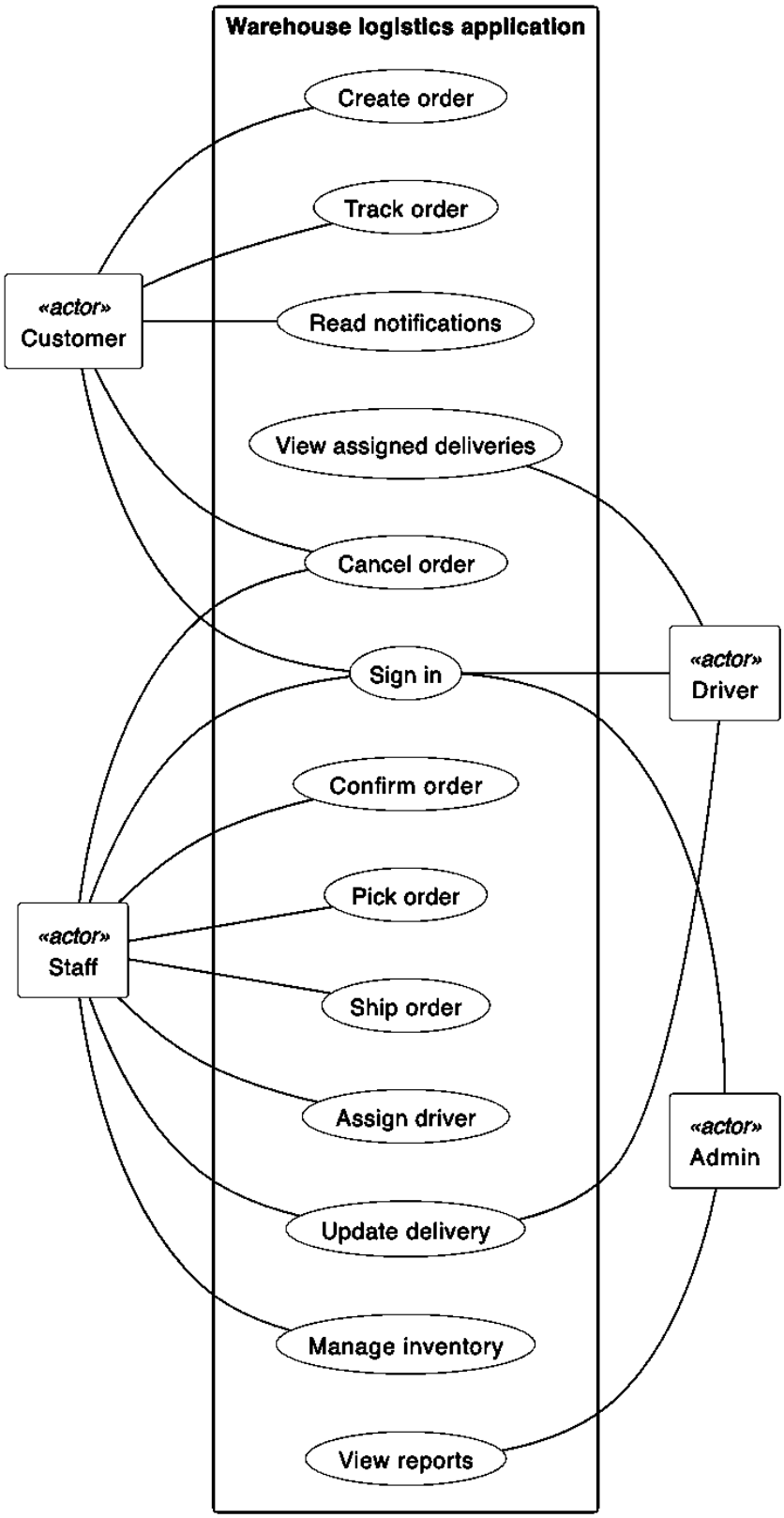


Figure 1. Role functions. Source.

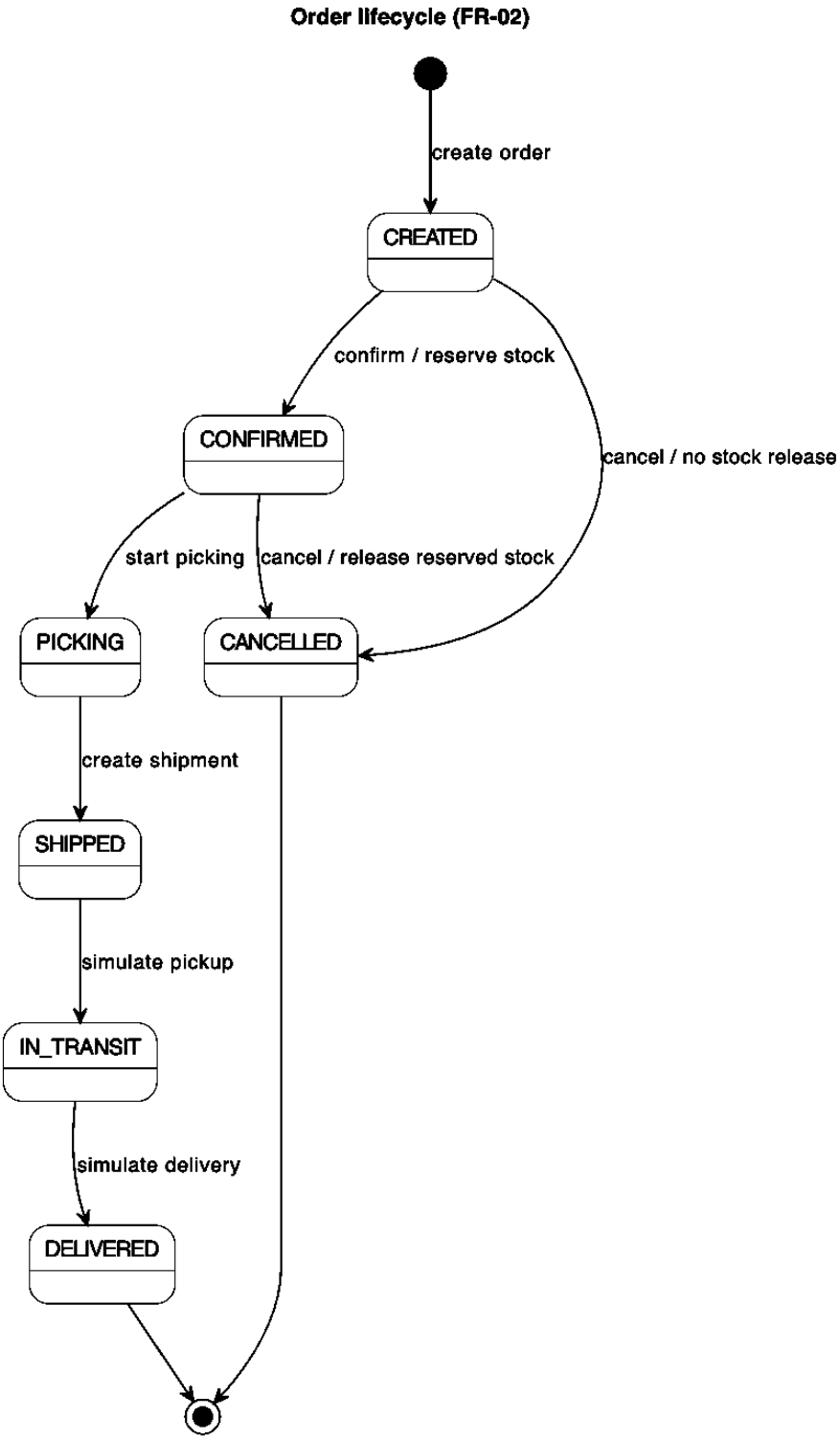


Figure 2. Order transitions. Source.

Each item references one product; each order requires at least one item; orders have 0..1 shipment; products have 0..* inventory logs.

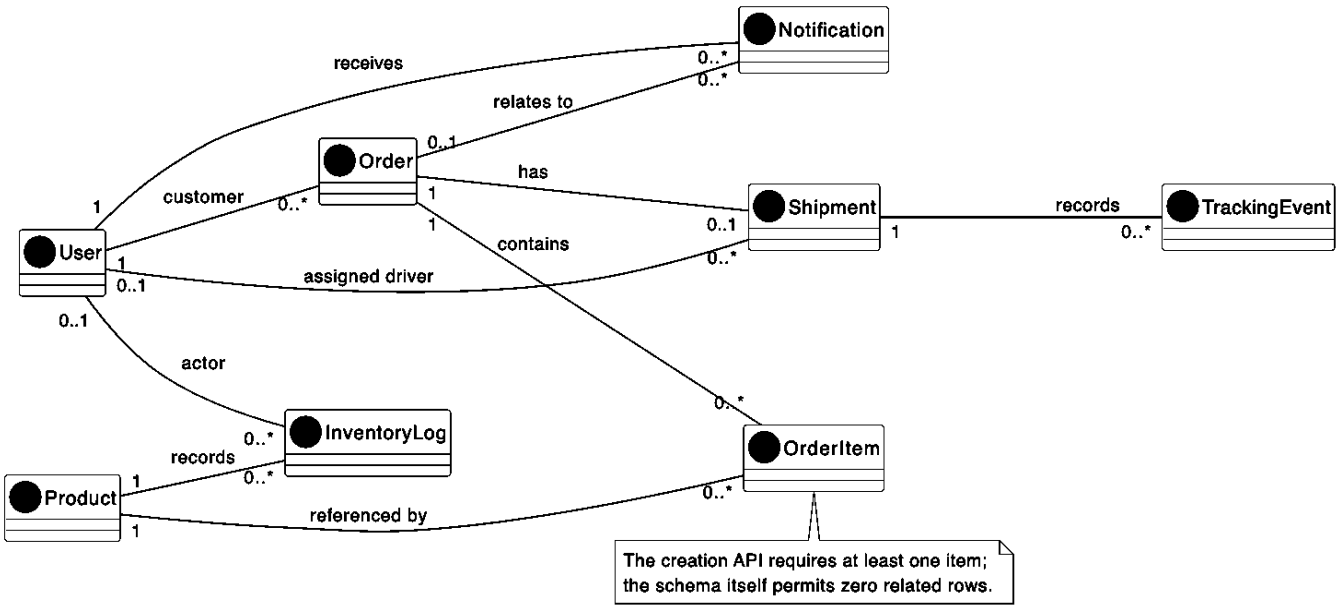


Figure 3. Data relationships. Source.

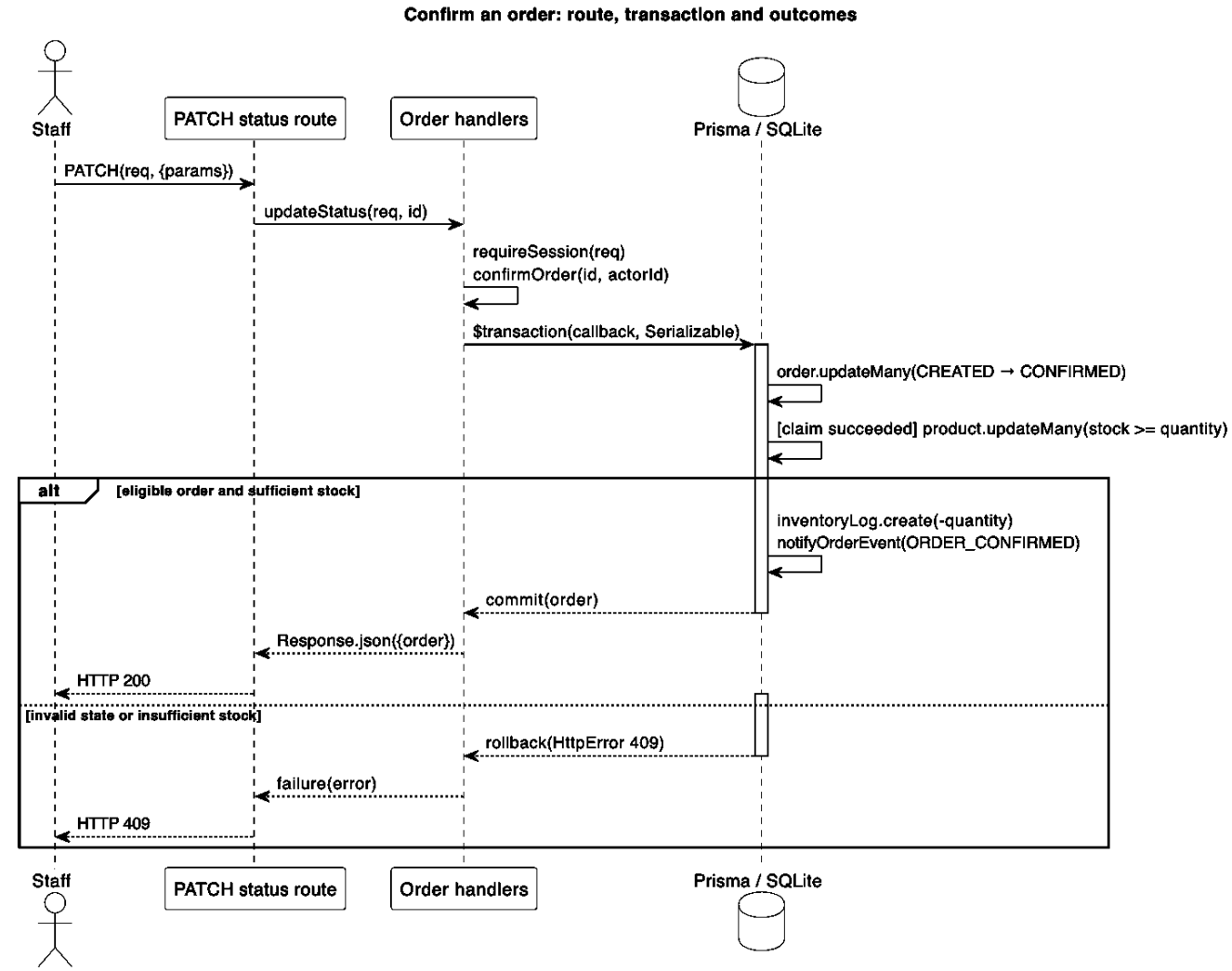


Figure 4. Confirmation transaction. Source.

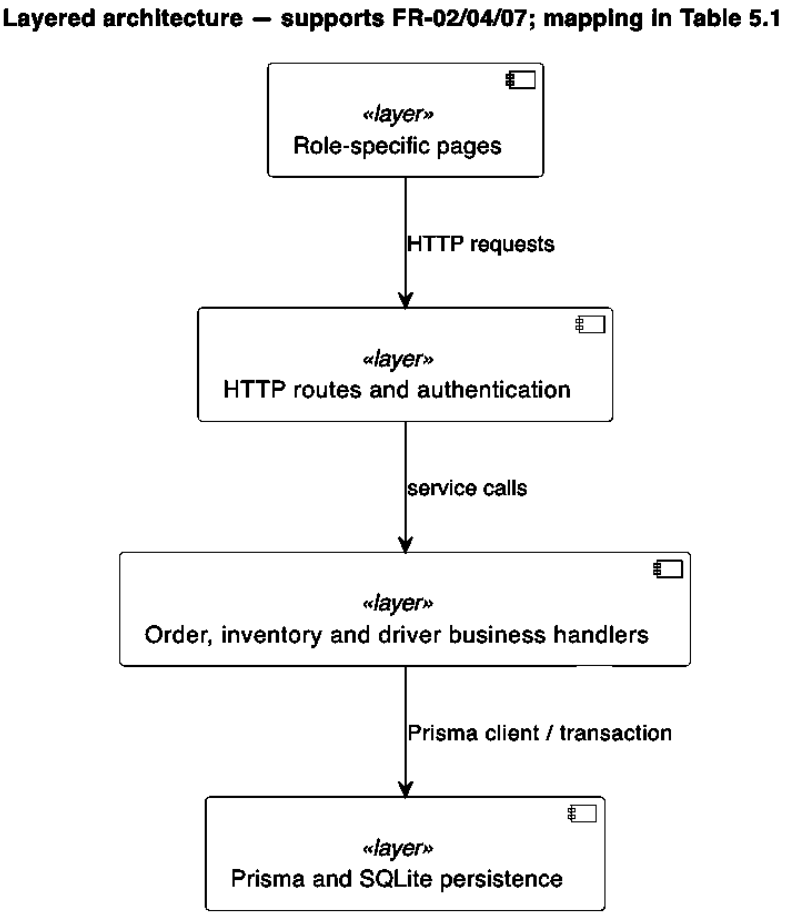


Figure 5. Four layers. Source.

5. Implementation

Stock is checked at creation and reserved at confirmation; the confirmation transaction (order-api.ts:245–293) enforces both. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED, with CANCELLED reachable from CREATED or CONFIRMED; cancellation per the matrix below. Shipments are created when staff ship.

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CONFIRMED	cancel, restock
Customer, staff	PICKING onwards	rejected

No cancellation after PICKING; orders proceed to delivery.

Design	Implementation	Verification
Guards: CREATED and sufficient stock	order-api.ts:250–253, 276–280	Orders
Confirmation transaction: a single transaction writing state, stock, ledger and notifications; rolls back on insufficient stock or repeated confirmation	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver
FR-03 numbering and tracking	create-order.ts:91, shipment-api.ts:62	Logistics chain
FR-04 inventory ledger	order-api.ts:281, 398	Inventory

6. Verification and validation

Verification checks requirements against the implementation; validation exercises the delivered workflows through the interface (§8). The regression log and CI show 38 unit, 140 integration, 2 delivery checks (delivery checks), and 26 Playwright browser checks.

In a local-loopback measurement on 11 November 2026, API p95 was 3.2–13.6 ms across endpoints and first-contentful-paint p95 was 64–96 ms, against targets of at most 2 s and 3 s. NFR-01 is met. NFR-02 is evidenced by recorded green build, test and browser runs (observations).

7. Evolution

The 17 Oct 2026 test-led review — defects collected by exercising the running app — found a confirmation endpoint that accepted repeat submits and a missing stock ledger; the confirmation transaction and its retest followed (26–28 Oct). The 10–11 Nov documentation review corrected the diagrams and aligned traceability with the corrected models. Two CI-health regression tests added on 11 Nov, after the regression exposed the gap, brought the suite to 38 (unit.log).

8. Validation

The automated walkthrough exercises eleven core operations — create order; order details; confirm; pick; ship; assign; pickup; delivery; report filter; export report; selected-date export — each performed by its owning role, at most three actions each; the walkthrough's 26 Playwright browser checks cover the eleven operations at one to three checks each, with expected outcomes asserted in the script. Results are in the walkthrough record.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagram bases diverged from the implemented schema	Models corrected	10–11 Nov documentation review	Diagra
Stock effects were recorded under FR-04	Moved to FR-02	17 Oct test-led review	Traceal
Trigger and outcome detail was thin	Expanded	10–11 Nov documentation review	Registe
The admin export took four scripted interactions	Simplified to three actions and regression-tested	11 Nov automated walkthrough	Issue #
CI health checks had no regression test	Suite reached 38	11 Nov regression work	Test his
NFR criteria assumed a hosted deployment model	Set for local deployment	10–11 Nov documentation review	Require

10. Evidence

Traceability · Baseline record — the recorded 12 Nov suite.